

DATA 557: Applied Statistics and Experimental Design

Final Project; Cape Town group

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Please find attached our group's final project submission for DATA 557. Our project investigates potential gender differences in faculty salaries at a U.S. university using the provided longitudinal dataset covering the years 1976–1995. Our submission includes two primary deliverables:

- Interactive Data Story (Python Jupyter/Colab Notebook)
- Case Study Analysis

Together, these components address the project questions using statistical modeling, visual exploration, and written interpretation. Below we summarize the deliverables, associated coding files, and the contributions of each group member.

All analysis and code files, as well as project deliverables can be found at:

https://github.com/DowntonCrabby/Data557_GroupProject_CapeTown

Interactive Data Story

Contributors: Sophia Koehn, Clark Roll, Katelyn Vu

Questions Addressed:

Q2: Does wage discrimination exist in granting promotions from Associate Professor to Full Professor?

Description:

The notebook presents the analysis as a step-by-step walkthrough combining statistical models, visualizations, and interpretation. The analysis evaluates promotion outcomes using three complementary approaches.

Associated Analysis Files:

- FinalProject_Q2_DataStory (python jupyter/colab notebook)
- FinalProject_DatasetCleaning (python jupyter/colab notebook)
- cleaned_longitudinal_dataset.csv

Individual Contributions:

Clark Roll

- Generation of dataset cleaning notebook, Prepared the analytical dataset used in the data story notebook
- Implemented the first two analytical approaches and interpretation of the statistical results
 - Logistic regression examining probability of promotion.
 - Linear regression examining time to promotion.
- Cover letter

Sophia Koehn

- Final summary of the interactive story
- Implemented visualizations and interpretation of plots:
 - Promotion differences by sex – box plot
 - Promotion differences vs salary by sex - scatter plot
 - Number of men and women who were promoted – bar graph

Katelyn Vu

- Subsetted data to perform visualizations of promotions difference
 - Created line plots and produced analysis for this subset
- Implemented linear regression to see effects of sex on the difference in addition to salary after promotions
 - Produced analysis for outcomes

Case Study Analysis

Contributors: Fatima Fazil, Priyal Jain, Amber Yang, Katelyn Vu

Questions Addressed:

Q1 Does wage discrimination exist in the starting salaries of faculty members (i.e., salaries in the year hired)?

Q3 Overall, how would you answer the question: Is there wage discrimination in salaries at the university?

Q4 What issues are involved in generalizing your results?

Description:

This case study analysis focuses on analyzing potential gender-based disparities among university faculty by applying hypothesis testing, regression modeling, and permutation methods to analyze differences in starting salaries, promotions, and overall compensation.

Associated Analysis Files:

- Q1_final(with permutation).R

- Q3_final.R

Individual Contributions:

Fatima Fazil

- Code, analysis and write up for question 3

Priyal Jain:

- Code, analysis and write up for question 4

Amber Yang

- Code for permutation test, data analysis and write up for question 1

Katelyn Vu

- Produced the code for:
 - Mean comparisons boxplots by sex
 - Mean comparison boxplots by sex and year
 - Linear regression model for the starting salary differences by sex